ENGINEERING ECONOMICS AND FINANCIAL MANAGEMENT

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DEPARTMENT VISION

The Department of Computer Engineering strives to be a center of academic excellence in computer engineering education, fostering multidisciplinary research and empowering individuals to embrace global opportunities.

DEPARTMENT MISSION

- Provide strong conceptual knowledge and hands-on experience to empower students for developing solutions by applying design thinking using state-of-the-art tools and technologies to shape individuals into successful professionals.
- Foster a dynamic research and innovation ecosystem that drives multidisciplinary advancements in computing.
- Inculcate professional ethics and effective communication skills to nurture leadership potential to address challenges in the field of Computer Engineering.

ENGINEERING ECONOMICS AND FINANCIAL MANAGEMENT

- Course Code: 1309101
- Credit/s: 01
- Theory: 01 hr/week
- Prerequisite: Basic Mathematics, Engineering concepts
- CIE: **25 Marks**
  - Attendance: 05 marks
  - Activity Based Learning: 20 marks

GRADIENT CASH FLOW

- A Gradient Cash Flow occurs when cash inflows or outflows change each period by a fixed amount or fixed percentage.
- 1. Linear/Arithmetic GS: Sometimes, cash flows will increase or decrease by a set amount, *G*, the gradient amount. This type of series is known as a **strict gradient series**, as seen in Figure
- 2. Geometric Gradient Series: the series in a cash flow is determined, not by some fixed amount like $500 but by some *fixed rate* expressed as a percentage.

1. LINEAR GRADIENT CASH FLOW

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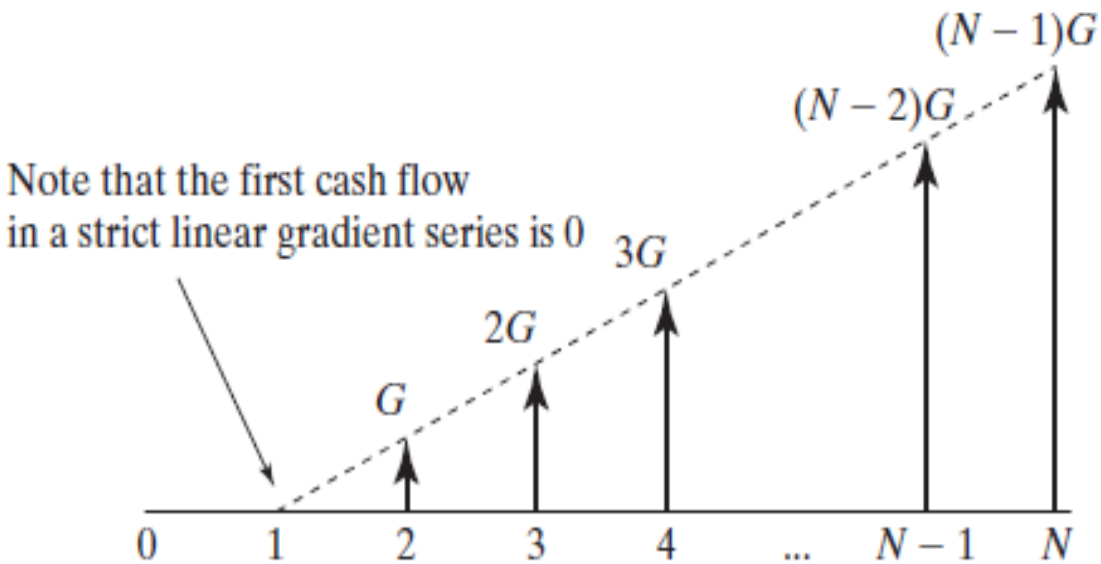


Cash flow diagram of a strict gradient series.

1. LINEAR GRADIENT CASH FLOW

- To find **gradient-series present-worth factor P**

$$P = 0 + \frac{G}{(1+i)^2} + \frac{2G}{(1+i)^3} + \cdots + \frac{(N-1)G}{(1+i)^N},$$

or

$$P = \sum_{n=1}^{N} (n-1)G(1+i)^{-n}.$$

After some algebraic operations, we obtain

$$P = G\left[\frac{(1+i)^N - iN - 1}{i^2(1+i)^N}\right] = G(P/G, i, N).$$

1. LINEAR GRADIENT CASH FLOW

- To find an equal-payment series equivalent to the gradient series, (**gradient-to-equal payment series conversion factor)** **A**

$$(A/G, i, N), \text{ or } A = G\left[\frac{(1+i)^N - iN - 1}{i[(1+i)^N - 1]}\right].$$

LINEAR GRADIENT SERIES AS COMPOSITE SERIES

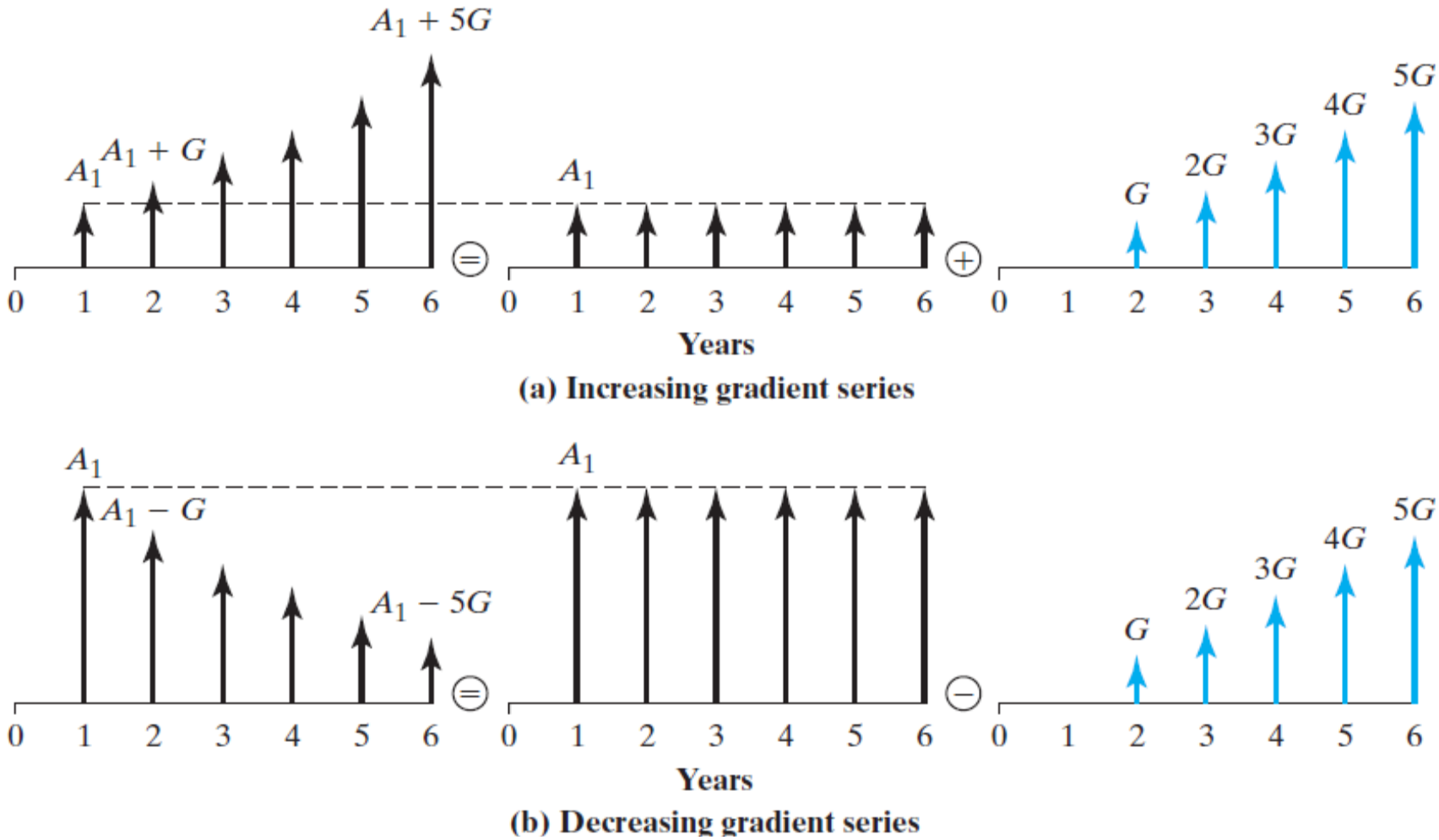
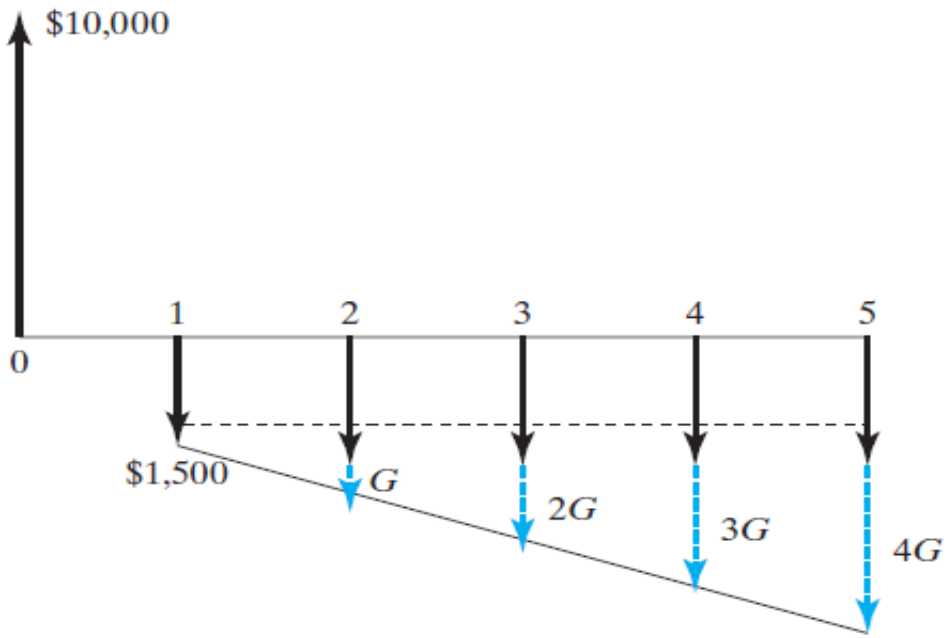


Figure 2.27 Two types of linear gradient series as composites of a uniform series of N payments of A_1 and a gradient series of increments of a constant amount G.

Ex: Creating a graduated loan repayment with a linear gradient series

- You borrowed $10,000 from a local bank with the agreement that you will pay back the loan according to a graduated payment plan. If your first payment is set at $1,500, what would the remaining payment look like at a borrowing rate of 10% over five years?
- Basically, we are calculating the amount of gradient (G) such that the equivalent present worth of the gradient payment series will be exactly $10,000 at an interest rate of 10%.
- **Given:** P = $10,000, $A1$ = $1,500, N = 5 years, and i = 10% per year.
- **Find:** G.

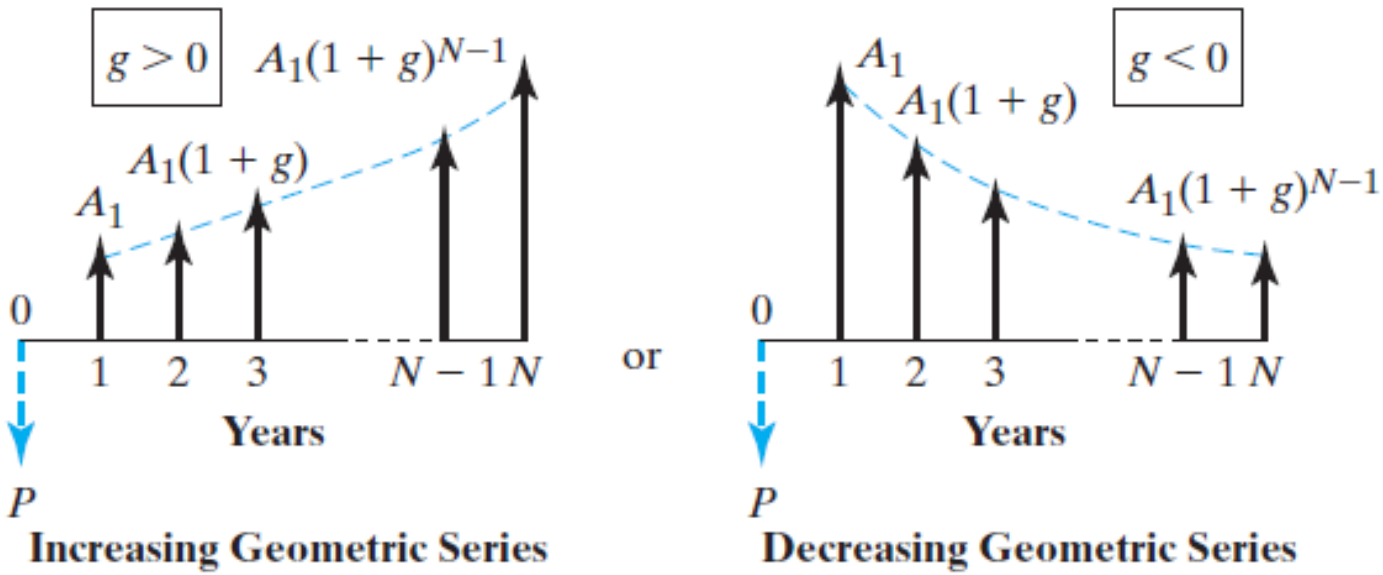


Creating a graduated loan repayment with a linear gradient series

- *Method: Calculate Present Value*
- Since the loan payment series consists of two parts—(1) a \$1,500 equal-payment series and (2) a strict gradient series (unknown, yet to be determined)—we can calculate the present value of each series and equate them with \$10,000:
- $\$10{,}000 = \$1{,}500(P/A, 10\%, 5) + G(P/G, 10\%, 5)$
- $= \$5{,}686.18 + 6.8618G$
- $6.8618G = \$4{,}313.82$
- $G = \$628.67.$

2. HANDLING GEOMETRIC GRADIENT SERIES

- Construction costs or maintenance costs, involve cash flows that increase or decrease over time by a constant percentage (**geometric**), a process that is called **compound growth**
- Price changes caused by inflation are a good example of such a geometric series.



2. HANDLING GEOMETRIC GRADIENT SERIES

- **Present-Worth Factor: Find *P*, Given *Ai*, *g*, *i*, and *N***
- we apply the **single-payment present-worth factor**

$$P = \sum_{n=1}^{N} A_1 (1+g)^{n-1} (1+i)^{-n}.$$

$$P = \begin{cases} A_1 \left[\dfrac{1-(1+g)^N (1+i)^{-N}}{i-g} \right], \text{if } i \neq g; \\ A_1 \dfrac{N}{(1+i)}, \text{if } i = g \end{cases}$$

- Or we can write $P = A1(P/A1, g, i, N)$

REFERENCE

- “ Fundamentals of Engineering Economics”, by C. S. Park Copyright © 2005
- “Contemporary Engineering Economics”, Fourth edition by Chan Park